

Bachelor's Programme in International Business

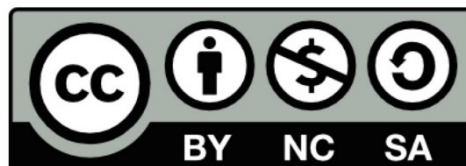
Does Website Optimization Influence AI Accuracy? The Role of Answer Engine Optimization

Summarizing Financial Figures from U.S. Listed Companies

Anton Kultavirta

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Title of thesis Does Website Optimization Influence AI Accuracy? The Role of Answer Engine Optimization

Programme Bachelor's Programme in International Business

Major International Business

Thesis supervisor Prof. Henrik Dellestrand

Collaborative partner IDX Inc.

Date 13 April 2026 **Number of pages** 53 +(6) **Language** English

Abstract

This thesis provides the first indication for the relationship between Answer Engine Optimization (AEO) and AI accuracy for AI-generated quarterly earnings summaries. Given the rapid development of AI models and increasing investor use of such tools, inaccurate AI-generated figures that may arise from poor digital presence could misrepresent a company's financial position to investors. A quantitative approach was implemented, using a sample of 105 companies listed on major U.S. stock exchanges to statistically examine the relationship. The significance and the influence of specific AEO areas are explored, with schema markup and content structure emerging as the most influential predictors of accuracy. Implications for listed companies and service providers are discussed, while the findings establish a foundation for further research.

Keywords AEO, GEO, AIO, AI, answer engine optimization, generative engine optimization, ChatGPT, artificial intelligence, AI accuracy, AI-generated summary, financial reporting, quarterly earnings, investor relations, large language models, hallucination, IR websites

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Työn nimi Does Website Optimization Influence AI Accuracy? The Role of Answer Engine Optimization

Koulutusohjelma International Business -kandidaattiohjelma

Pääaine International Business

Työn valvoja Prof. Henrik Dellestrand

Yhteistyötaho IDX Inc.

Päivämäärä 13.4.2026

Sivumäärä 53 +(6)

Kieli englanti

Tiivistelmä

Tämä kandidaattityö tarkastelee yhteyttä vastauskoneoptimoinnin (answer engine optimization, AEO) ja tekoälymallien vastausten tarkkuuden välillä, tarjoten ensimmäisen osoituksen niiden välisestä suhteesta tulosjulkistusten yhteydessä. Tekoälymallien nopea kehitys sekä niiden kasvava hyödyntäminen sijoittajien keskuudessa avaa mahdollisuuden vääristyneisiin tuloslukujen tarjontaan tekoälymallien toimesta. Tämän voidaan osittain katsoa johtuvan heikosta verkkosivujen optimoinnista, mikä voi johtaa yhtiön valheelliseen edustamiseen sijoittajien keskuudessa. Kvantitatiivinen menetelmä otettiin käyttöön yhdessä 105 yrityksen näytteellä. Näyte koostuu yrityksistä, jotka ovat listautuneet Yhdysvaltojen suurimpiin pörssihin. Näyte mahdollistaa yhteyden tarkastelemisen tilastollisten menetelmien avulla. Tilastollista merkitystä ja vastauskoneoptimoinnin eri alueiden vaikutusta tulkitaan. Rakenteellinen data sekä sisällön rakenne nousevat tärkeimmiksi muuttujiksi ennustimien suhteen. Seurauksia pörssilistautuneille yrityksille ja palvelun tarjoajille tarkastellaan, tarjoten samalla perustan tuleville tutkimuksille aiheesta.

Avainsanat AEO, GEO, AIO, AI, answer engine optimization, generative engine optimization, ChatGPT, artificial intelligence, AI accuracy, AI-generated summary, financial reporting, quarterly earnings, investor relations, large language models, hallucination, IR websites

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Preface and acknowledgements

First, I want to thank my thesis supervisor, Henrik Dellestrand, for his guidance, advice and vision for the thesis structure. Likewise, I am grateful for the guidance, materials and support provided by IDX Inc.'s staff, Jukka Ylikitti, Liron Hadar and Stefan Bardega. Their initial feedback and the auditing tool by Bardega were essential to completing the study within a demanding schedule.

Lastly, I want to thank my family and friends for their interest and support throughout the past few months, which sparked motivation to move forward.

Mikkeli, 13 April 2026
Anton Kultavirta

Symbols and abbreviations

Symbols

n	sample size
p	p-value
SD	standard deviation
r	Pearson correlation coefficient
r_s	Spearman correlation coefficient
D	rank difference
t	t-statistic
$\bar{d}$	mean difference
S_d	standard deviation of differences
B	unstandardized regression coefficient
β	standardized regression coefficient
β_0	intercept
F	F-statistic / Significance F
R^2	coefficient of determination
R_a^2	adjusted R squared
X_1, X_2, X_3, X_4	independent variables
Y	dependent variable
ϵ	error term
e	margin of error
N	total population

Operators

$\times$	multiplication
$+$	addition
$-$	subtraction
$/$	division
$<$	less than
$=$	equals
Σ	summation
$\sqrt{\quad}$	square root
$\quad^2$	squared
$\%$	percentage

Abbreviations

Acc	accuracy
AEO	answer engine optimization
AI	artificial intelligence
EBITDA	earnings before interest, taxes, depreciation and amortization
E-E-A-T	experience, expertise, authoritativeness and trustworthiness
ESMA	European Securities and Markets Authority
EU	European Union
GAAP	Generally Accepted Accounting Principles
GPT	generative pretrained transformer
H	hypothesis
HTTPS	Hypertext Transfer Protocol Secure
IFR	internet financial reporting
IFRS	International Financial Reporting Standards
IR	investor relations
JSON	JavaScript Object Notation
JSON-LD	JavaScript Object Notation for Linked Data
Mdn	median
ML	machine learning
iXBRL	inline eXtensible Business Reporting Language
PLC	publicly listed company
SEO	search engine optimization
URL	uniform resource locator
U.S.	United States of America
vs.	versus

Definitions

Answer Engine	A system designed to provide direct answers to user queries, commonly facilitated by AI.
Answer Engine Optimization	The practice of structuring online content for easy and authoritative interpretation, extraction and presentation by AI answer engines. This broadly includes all company disclosures such as management strategy and sustainability, while emphasizing machine-readability and citability. For financial information, a supplementary aspect is increasing the probability that AI outputs reflect official disclosures from the source origin (IR website).
Artificial Intelligence	Algorithmic (automated set of instructions) systems created to perform tasks that historically required human cognitive skills.
Citability	The AI's ability to select the most relevant and authoritative source for a response when filtering through a collection of online sources.
Content Structure	The organization and formatting of website content to improve readability and interpretation by answer engines.
Database	An organized collection of stored information.
E-E-A-T	A framework for evaluating the quality of online content across four areas: Experience, Expertise, Authoritativeness and Trustworthiness.
Hallucination	When AI generates information that is incorrect or not based on factual sources.
Input	A given request or inquiry to a system.

Investor Relations	The practice of maintaining transparent information flow between a company and its investors, typically through financial reports and other disclosures.
Large Language Model	A type of AI model trained on large amounts of text to generate human-readable responses.
Machine-readability	The ease of AI systems in interpreting and extracting online information for use in generating responses.
Neural Networks	A system of automated processes designed to interpret and provide information, while mimicking the way the human brain works.
Output	A system's response to a given request.
Publicly Listed Company	A company whose shares are publicly traded on a stock exchange.
Quarterly Report	A financial report published every three months by listed companies.
Schema Markup	Structured code ("label") added to a website to help answer and search engines interpret its content.
Structured Data	Information organized into a format, such as rows and columns, making it easily readable for machines.
Technical SEO	The optimization of a website's technical infrastructure, such as speed and layout, to improve its accessibility and interpretation by search and answer engines.
Web Crawlers	Automated programs that periodically browse web pages to discover, interpret and index information into a database.
Zero-shot Prompting	A prompting technique where a model receives no prior examples before responding to an inquiry.

1 Introduction

This study provides empirical evidence on the relationship between Answer Engine Optimization and the accuracy of AI-generated quarterly report summaries in early 2026. Historically, the emergence of financial reporting that fosters investor decision-making today can be traced to 18th century France (Tache & Costea-Marcu, 2021). Notably, accounting standards and related legislation have progressed in the early 21st century (Ball, 2006; Soderstrom & Sun, 2007). However, recent advancements in artificial intelligence (AI) have raised new questions for both investors and publicly listed companies (PLCs) regarding the summarization of financial reporting. As investors increasingly utilize AI to summarize and interpret company financials (Blankespoor et al., 2026; Oyeniyi et al., 2024), the accuracy of these outputs has yet to be empirically evaluated. This is particularly relevant for understanding to what extent a firm’s digital presence, including its online disclosures, may be optimized for generated answers. The practice in question is known as Answer Engine Optimization (AEO). Whether higher AEO corresponds to greater accuracy in AI-generated financial outputs for investors remains an open question.

A fundamental purpose of financial reporting is to provide a reliable basis for security and firm valuation (Holthausen & Watts, 2001; Kothari, 2001). While standardized accounting frameworks, such as IFRS and GAAP, aim for neutrality and error-free disclosure (IFRS, 2018; McCallig et al., 2019), the reports are often technically complex (Lesmy et al., 2025). This complexity increases the risk for a higher knowledge gap between PLCs and investors (Bergh et al., 2018; Guay et al., 2016). Historically, the lack of technical competence and access to the latest technology has created barriers that have prevented certain investors from escaping a continuous cycle of market disadvantage (Sias & Starks, 1997; Inci, 2023, p. 52). However, the mass adoption of AI with Large Language Models (LLMs) have equalized opportunities in recent years. Yet, it has introduced a new risk with algorithmic opacity. Since AI models function as “black boxes” sacrificing output transparency (Mühlbacher et al., 2014; Ribeiro et al., 2016; Riveiro & Thill, 2022) through hidden layers of processing (Goz et al., 2019; Jiménez-Pérez et al., 2023; Singh et al., 2019), they are prone to critical errors and fabricating information (Granstedt et al., 2025; Lee et al., 2025; Sood et al., 2025; Tirumala et al., 2024).

AI models such as ChatGPT, Gemini and Claude increasingly act as intermediaries between PLCs and investors (Breuer et al., 2025; Blankespoor et al., 2026). Simultaneously, the models prioritize accessible, structured, citable and authoritative information (Chen et al., 2025; Smajić et al., 2025). As PLCs differ in their adoption of digital optimization, investors may receive vastly different representations of the same firms. This raises concerns regarding the consistency and reliability of AI-generated financial summaries. Such concerns are especially present when these outputs are used to support investment decisions (Blankespoor et al., 2026; Yeo et al., 2025). Notably, Google's organic click-through rate has reportedly declined by 65.2 percent year-on-year (McDonald, 2025), which further emphasizes PLCs' potential incentive to improve online representation through AEO to benefit from accurate AI visibility. Firms that are not highly optimized may encounter instances where their investors receive fabricated information through AI. These instances pose a challenge, since disclosures are no longer limited to text or files published on websites but now also include AI-generated summaries. PLCs are therefore required to communicate with investors through both channels.

PLCs periodically publish financial disclosures as signals to investors. A primary and most trusted channel for these signals is the official investor relations (IR) website (Kelton & Yang, 2008; Nel & Baard, 2022). Despite the costs and possible market reactions from reporting, information asymmetry remains (Bergh et al., 2018). When AI intermediaries distort the information flow through hallucinations, the knowledge gap between PLCs and investors may be exacerbated. This weakens the core purpose of disclosures. As a response, AEO relies on optimizing online content for machine-readability and citability to AI systems. While AEO broadly extends to all company disclosures, this study focuses specifically on financial figures. Arguably the foundational pillars include four areas: from using identifying codes that label website content (I. Schema markup), to having organized sections, short paragraphs and unambiguous wording (II. Content structure). It further requires website accessibility (III. Technical SEO), alongside Google's Experience, Expertise, Authoritativeness and Trustworthiness (IV. E-E-A-T) framework (Chen et al., 2025; Daoud et al., 2024; Google 2025a; Google 2025b; Schema markup.org, 2026).

The risk of inaccuracies in summaries also has economic consequences. Successful implementation of standards such as IFRS has shown to result in higher trading frequency and greater stock price stability (Sultan et al., 2024). However, if AI intermediaries distort the information flow, the benefits of transparency may be reversed. Inaccuracies in reported figures may reduce the usefulness of financial information for decision-making (Dechow et al., 2010; Palmrose et al., 2004). If the PLC is perceived as unreliable by the market and investors, financial lenders may impose higher interest rates and collateral requirements. The consequences appear consistent with the negative effects when transparency is compromised (Chen et al., 2015). This creates a puzzle in which firms must comply with standards to the best of their abilities to ensure accuracy (IFRS, 2018), yet they appear to have no direct control over the “black box” that summarizes their results for investors.

Currently, academic literature lacks empirical studies concentrated on this phenomenon. Specifically, no prior work has examined whether AEO may mitigate hallucinations and improve accuracy. Therefore, the following research question arises: *What is the relationship between a company’s Answer Engine Optimization and the accuracy of figures in AI-generated financial summaries?* This study aims to contribute to the early stages of AEO literature through a combination of primary and secondary data. The relationship is explored through the theories of algorithmic opacity, signaling theory and information asymmetry. The aim is to measure how accurate AI-generated summaries are by comparison with official PLC reports, while assessing differences in AI summary accuracy between pre-earnings and post-earnings. Similarly, AEO areas and their individual influence on accuracy are measured. The primary objective is therefore to examine the relationship between overall AEO score and accuracy, where accuracy refers to error-free alignment with official disclosure figures reported by PLCs.

To address the research question, a sample of 105 companies listed on major U.S. stock exchanges is utilized alongside a widely adopted ChatGPT model to generate financial summaries. Disclosures from two of the latest quarterly periods provide the basis for comparison, which results in accuracy scores for each company. The accuracy of summaries is assessed against audited AEO scores. The findings reveal that among the four AEO areas, schema markup and content structure emerge as the most influential predictors. Meanwhile, structured data implementation appears to be in its early stages within standard investor relations practices.

2 Literature review

2.1 Purpose of Financial Reporting

For a listed company to periodically share inside information: non-public information with significance to pricing (EU Regulation, 596/2014), the Internet and company websites in particular currently facilitate this as direct communication channels to investors (Boulland et al., 2026). This development is important because financial reporting provides the basis of information for security and firm valuation (Holthausen & Watts, 2001; Kothari, 2001).

2.2 Regulatory Environment: IFRS and GAAP

A major standardization of financial reporting in the EU began in 2005 with the adoption of IFRS, which was to be implemented by all member states. This was in accordance with regulation passed three years earlier by the European Parliament and the Council of the EU (EC, 1606/2002). Since then, the adoption of IFRS has accelerated. IFRS reports that 169 legal jurisdictions, primarily nations, have either adopted the standards or are considering them (IFRS, 2026). This development has prompted arguments both for and against the standardization of financial reporting.

In a study on the pros and cons of IFRS, Ball (2006) emphasizes that IFRS, as a principles-based system, relies on general rather than detailed rules. Because IFRS does not provide defined rules for all possible accounting scenarios, it has been argued to be incomplete in its coverage. This increases the need for accounting judgement by auditors, thereby raising the potential for inconsistency when comparing PLCs' financial reporting (ibid; Brown, 2013). In contrast, GAAP is acknowledged as closer to a rules-based system, providing detailed guidance for specific transactions and financial reporting (Bradbury & Schröder, 2012). Although the framework may not perform as initially intended, arguments supporting the implementation of IFRS have emerged.

For instance, in an IFRS literature synthesis, Yu (2022) argues for IFRS benefits such as PLC comparability. This allows investors to evaluate similarities and differences between firms while improving the reliability of reported financial figures. These advantages are said to outweigh the disadvantages highlighted by Ball (2006) and Brown (2013). Sultan et al. (2024) further support this with a study of 1,200 PLCs across 30 countries over a ten-year period (2010-2020). They find higher trading frequency, more unified investor sentiment reflected in lower bid-ask spreads and greater stock price stability following IFRS implementation. However, Chen et al. (2015) report that the increased transparency and comparability negatively

affected PLCs with higher interest rates and collateral requirements from bank lenders.

Taken together, the literature suggests that IFRS implementation can help PLCs meet investor expectations through improved transparency and comparability. Yet serious considerations remain for PLCs, particularly the initial costs and demands of IFRS adoption, which may become limiting factors when adoption is not mandated. This view is consistent with suggestions that IFRS benefits depend heavily on incentives and enforcement (Holthausen, 2009), with firms tending to avoid costly changes unless obligated (Kvaal & Nobes, 2010).

2.3 Company Perspective

It must be noted that PLCs are excluded from earning additional funding from publishing financial reports, after which share prices adjust (Botosan, 1997). Consequently, the information value of financial reports is arguably more relevant to investors than to the firms themselves. Nevertheless, PLCs benefit financially and reputationally from sharing information with investors (Healy & Palepu, 2001; Simpson & Tamayo, 2020).

Generally, cost of capital refers to the expected return that investors are able to earn on the best alternative investment with similar duration and risk (Berk et al., 2024, p. 286). More intuitively for PLCs, it is the return demanded by shareholders and financing lenders. Lambert et al. (2007), in developing a cost of capital model, report that accounting information derived from financial reports can lead to a lower cost of capital for firms. Specifically, it is argued to alter perceptions of future cash inflows, which may lead the firm to be perceived as having a lower risk of bankruptcy. However, empirical testing has left the link between financial reporting and cost of capital unclear and statistically insignificant (Mohanram & Rajgopal, 2009). Rather, the extent of voluntary disclosure may be more relevant for firms' financing costs (Botosan, 1997), as perceptions of transparency can influence required returns (Francis et al., 2008).

Voluntary disclosure can contribute to a firm's reputation for transparency (Healy & Palepu, 2001). However, PLCs may also benefit financially from publishing financial reports. Baker and Wurgler (2002) argue that PLCs are most likely to issue new shares when market value ($\text{Shares Outstanding} \times \text{Stock Price}$) is higher than past or reported estimates in financial reports. They also note that firms tend to repurchase their own shares at lower prices. This theory of equity market timing, when considered alongside financial reporting, suggests that reputational interest can raise a firm's share price to an all-time high (Raithel & Schwaiger, 2015). In such a case, a PLC in need of additional capital could issue shares more favorably. A higher market

valuation would allow the firm to raise the required capital while ceding less ownership in shares.

Although this theory has been regarded as valid in principle (Huang & Ritter, 2009), Myers and Majluf's (1984) pecking order theory suggests that firms issue additional shares out of necessity due to insufficient funds rather than to benefit from market timing. In summary, PLCs do not necessarily publish financial reports to obtain direct financial benefits. Nevertheless, the information they provide may create opportunities for growth and financing (Simpson & Tamayo, 2020).

2.4 Investor Perspective and Information Asymmetry

Investment opportunities appear for investors as PLCs share information periodically (Huang et al., 2025). These disclosures reveal a gap between public and private information, as firms hold knowledge that may influence security pricing. This difference is referred to as information asymmetry (Bergh et al., 2018). For investors, this gap may present the primary risks of mispricing and potential loss of invested capital (Buckley et al., 2012; Li, 2020).

One of the foundational perspectives on how investors value financial reporting comes from the liquidity-based view of Amihud and Mendelson (1986). Their argument is based on the principle that investors require compensation for bearing risk. In this context, when a PLC's shares are highly liquid, referring to ease of purchasing and selling, investors demand a lower return due to reduced trading-related risk. A lower required return indicates a higher share price, allowing PLCs to issue additional equity at more favorable prices. This allows firms to raise more capital while surrendering a smaller share of ownership. The higher share price, in theory, is consistent with valuation models such as the Gordon Growth Model (1962), where future cash flows are discounted at a lower rate due to reduced risk. Frequent and transparent financial reporting by PLCs has also seen an increase in trading volume (Qian et al., 2025). From an investor perspective, higher liquidity may attract short-term investors as a larger pool of buyers can provide more exit opportunities.

While earlier research established a link between trading activity, liquidity and the cost of capital for PLCs, other studies have questioned the strength of this relationship. A central concern relates to how information asymmetry influences investor behavior in the context of financial reporting. This occurs when one party possesses superior information, creating a knowledge imbalance (Bergh et al., 2018). In theory, high-quality reporting reduces uncertainty and may lower the return demanded by investors. However, Inci (2023, p. 52) notes that investors act on new information before prices fully adjust, increasing competition among market participants.

As such, although timely reporting can influence investor perceptions, they may not fully process all available information.

Diamond and Verrecchia (1991) provide theoretical support for this argument by showing that reduced information asymmetry from frequent disclosures increases market competition and reduces volatility. Furthermore, empirical evidence suggests that extensive disclosure can lead to higher financing costs. Botosan and Plumlee (2000) find that firms with more extensive disclosures experienced a higher cost of equity compared to firms providing only the required amount. Reporting quality and quantity therefore appear to shape investor expectations beyond financing costs.

The information asymmetry perspective is also supported by other investment literature. Biddle et al. (2009) show that higher quality mitigates both overinvestment and underinvestment by reducing the information gap between PLCs and external capital providers. Similarly, Gomariz and Ballesta (2014) and Verdi (2006) find that high-quality reporting limits overinvestment through improved monitoring, thereby reducing speculative behavior. However, this perspective overlooks the possibility that information itself may introduce risk rather than simply reduce it.

Easley and O'hara (2004) provide an opposing view in their investigation of information on cost of capital, while developing an asset pricing model. They argue that information risk is included within expected returns, such that investors that hold private information have an edge against uninformed investors holding the same stock. They view this risk as systematic, meaning investors cannot diversify away from it. This challenges the liquidity premium theory by suggesting that information uncertainty is a greater concern for investors than trading ease alone.

While literature on information asymmetry commonly supports high-quality reporting, some studies raise concerns regarding flawed investor decision-making. Gigler et al. (2014), in a study of financial reporting frequency, note that more frequent or detailed reporting increases the risk of misinterpretation. In turn, investors may prioritize short-term profitability over longer-term market stability. Additionally, empirical results from Impink et al. (2025) and Shields (1983) reveal that increased information reduces managerial accuracy in decision-making. This is due to information overload, where information processing becomes complex.

Overall, these perspectives do not fully account for the possibility that increased complexity in financial reporting can hinder the ease of identifying negative developments for investors (Aghamolla & Smith, 2023). This suggests that reporting complexity may influence how investors behave in financial markets. However, the extent of this influence may depend on the capabilities and characteristics of different investors, which guide market behavior (Daniel et al., 2002).

2.5 Market Behavior and Investor Decision-making

Investor behavior in the secondary market (i.e. stock market) is influenced by information published in financial reports, such as quarterly earning disclosures (Israeli et al., 2022). These markets are typically open for transactions five days a week, while remaining closed on weekends (DellaVigna & Pollet, 2009). Therefore, the broad category of investors is differentiated into institutional and retail investors. Generally, pension funds, insurance companies and mutual funds, among others, are referred to as institutional investors. These are entities managing vast quantities of capital to be invested in assets, including shares of PLCs (Berk et al., 2024, p. 871). On the other hand, non-professional individuals who participate in the secondary market, either by selling, purchasing or trading securities are referred to as retail investors (Kumar & Lee, 2006).

Under efficient market theory, where information is reflected in the prices of securities, secondary markets are considered semi-strong efficient when all publicly available information is taken into account (Inci, 2023, p. 122). A recent study on market efficiency by Lan et al. (2024) finds that the phenomenon of post-earnings announcement drift, referring to share price movements dictated by surprising information in financial reports, appears to continue in today's market. After all, the first discovery of such a phenomenon dates to the late 1980s by Bernard & Thomas (1989). These findings suggest a limitation to semi-strong efficiency, as prices reflect a lagged drift rather than an instant adjustment.

Earnings disclosures tend to reveal differences in how institutional and retail investors react. Historically, institutional investors have adapted faster and more dominantly to earnings disclosures, with their trading patterns recorded as dictating daily stock returns (Sias & Starks, 1997). This is consistent given their larger quantity of capital and the expectations of efficient market theory. Inci (2023, p. 52) equally notes the institutional use of hollow-core fiber cables, which allow transaction orders to travel close to the speed of light. The edge compared to retail investors appears prevalent. Despite this advantage, new studies on institutional behavior in the late 2020s

could yield unique insights and address the existing literature gap regarding technological developments.

A study on transaction data from the U.S. stock market by Barber et al. (2009) reports distinct patterns in retail investor behavior. Retail investors often appear to purchase the same stocks across months without major influence from financial reports on average. Yet, for stocks with a market value below two billion U.S. dollars (also referred to as small-cap stocks), retail investors tend to impact price movements more significantly. This tendency aligns with the popular investing strategy of dollar-cost averaging, where specific stocks are purchased with limited short-term information considerations (Brennan et al., 2005).

However, when information from financial reporting is considered, Kelley and Tetlock (2013) suggest that retail investor transactions may reflect what is observable of a firm rather than informed decisions. Supporting this, Hsieh et al. (2020) identify a pattern for retail investors to collectively herd in their investment decisions and argue that easily accessible information from search engines may contribute to such grouping. Furthermore, based on a survey by Brooks and Williams (2022), personal characteristics and demographics can drive retail investor decision-making, especially when volatility is higher than usual. Essentially, retail investor behavior based on financial reporting appears to lag behind that of institutions, which tend to process new information more rapidly. Such differences may also include the use of advanced technology and informed decision-making through a strict reliance on financial reporting and estimates derived from valuation models by institutional investors.

Additionally, there are arguments against the efficient market theory. Two notable pieces of evidence against public information being appropriately reflected in share prices are the size and January effects. Historically, stock market data suggests that small firms earn higher returns than larger firms even after risk adjustments, although the effect has slowed more recently (Fama & French, 1993; Fama & French, 2015; Hou et al., 2020). Moreover, smaller firms have historically gained abnormal returns in January compared to other months (De Bondt & Thaler, 1987; Haugen & Jorion, 1996). Yet questions have emerged regarding whether these recorded effects may instead be caused by other factors, such as selection bias (Herrmann & Hibbeln, 2021). While such anomalies have been reported, the theory appears useful for understanding how financial reporting influences market behavior.

2.6 Digitalization of Financial Information

The mass adoption of the internet in the 1990s accelerated the digitalization of financial information for PLCs. Initial benefits included lower reporting costs and more rapid communication of new information (Debreceeny et al., 2002; Ettredge et al., 2001). This development prompted PLCs to adapt to the changing environment through online disclosures, largely due to increased demand for transparency by investors and regulations (Christensen et al., 2021; Debreceeny et al., 2002; Directive 2013/50/EU).

2.6.1 Online Disclosures

Debreceeny et al. (2002) define internet financial reporting (IFR) as the distribution of financial and performance information via internet technologies. A primary and most trusted channel for IFR has been the official corporate investor relations (IR) website (Kelton & Yang, 2008; Nel & Baard, 2022), which firms can utilize to publish annual and quarterly results, financial statements, press releases, governance documents and other related disclosures. These websites provide investors with firsthand information about the firm, reducing the need for intermediaries (Halim et al., 2015). Despite their importance, IR websites often vary in disclosure quality. Differences in information accessibility and ease of navigation may hinder investors from obtaining information directly at its source (AbuGhazaleh et al., 2012; Halim et al., 2015).

Besides IR websites, PLCs rely on additional communication channels. Foremost, many firms utilize social media for announcements and other timely updates (Blankespoor et al., 2014). Notifications from email lists can similarly be distributed to subscribed investors, alongside responses to investor inquiries. For instance, calendar reminders, policy updates, result presentations and financial reports are typically communicated through these channels (Nel & Baard, 2022). Yet other channels for interactive two-way communication, such as earnings calls and investor conferences, exist between PLCs and investors to obtain information and insights directly from upper management in real time (Laskin, 2011). Additionally, many companies provide downloadable files, such as Excel spreadsheets that contain financial figures (Janvrin et al., 2013), although these are typically accessed via IR websites.

Furthermore, information formatting is also central to online disclosures. In 2021, the European Securities and Markets Authority (ESMA) mandated European PLCs to report their annual financial statements in a single electronic format and apply inline eXtensible Business Reporting Language (iXBRL) labels (Financial Supervisory Authority, 2025). These labels convert reported

figures into structured and machine-readable data (see [Appendix A](#)). Yet presentations remain human-readable (ESMA, 2025). Arguably, this change indicates a broader regulatory push toward structured and technologically verifiable disclosure, as ESMA (2025) considers applying similar labelling requirements to other PLC disclosures.

2.6.2 Utilization of Online Disclosures

Beyond official reports on IR websites, prior research has explored other forms of online disclosure, which vary in their effectiveness. In a survey study on PLC responsiveness, Nel and Baard (2022) find that PLC responsiveness to email requests is considerably low, at around half of the 77 companies sampled. This suggests that PLCs respond less frequently online when investors seek additional information without real-time interaction. Lee and Zhong (2022) note that PLCs consider the time and resources required to process and respond to inquiries, which can increase the cost of providing information.

Aligning with this perspective, Meirkulov et al. (2024) suggest that processing costs are reduced when PLCs focus on earnings calls, as these allow firms to address large groups of investors simultaneously. However, it is reported that during the question-and-answer section of these calls, around 30% of the questions included were unanswered in the related report or presentation. Additionally, Chen et al. (2018) observe a linear decline in both response times and the tone of management during these calls between 2002 to 2013. Rather than a decline in response quality, this trend may indicate a cautious approach to ensure all verbal disclosures align with factual information.

While PLCs arguably benefit from lower processing costs via online disclosures, the development of AI presents a new opportunity for firms to become more proactive in their investor communications. As AI can mediate investor access to PLC disclosures, structured data becomes relevant in potentially mitigating misinformation that may distort investor decision-making (Spitzer et al., 2025). Consequently, the responsiveness challenges highlighted by Nel and Baard (2022) may be partly addressed by optimizing corporate disclosures for AI systems through AEO practices.

2.7 AI, Large Language Models and Machine Learning

Fundamentally, Pomerol (1997) acknowledges artificial intelligence (AI) as the ability of computers and machines to receive, process and use information in ways that resemble human cognition. More recently, AI has been defined as the capability of algorithmic systems to perform tasks that

traditionally depended on human reasoning, learning, decision-making and pattern recognition (Collins et al., 2021; Gignac & Szoborai, 2024; Rattan et al., 2022). These algorithms refer to automated processes that follow instructions to solve problems without direct human involvement (Mahmud et al., 2024).

Rattan et al. (2022) recognizes machine learning (ML) as a branch of AI. Essentially, it refers to training systems to learn from data without detailed prior instructions (Brynjolfsson & McAfee, 2014 p. 28; Jarrahi, 2018). Within ML and AI, large language models (LLMs), such as ChatGPT, are trained on vast text data to generate human-readable text based on user prompts (Astobiza, 2025; Raza et al., 2025; Zhou & Chen, 2025). Specifically, generative pre-trained transformer (GPT) models are commonly used forms of AI and allow broad user access in applications such as financial information processing (Chitty-Venkata et al., 2023; Wang & Brorsson, 2025).

2.8 Artificial Intelligence in Financial Information Processing

In a recent empirical study, Wang and Brorsson (2025) evaluate the accuracy of LLMs in financial information processing. They test three models (Llama 3.1, Llama 3.2 and Mistral 7B) with different prompting and optimization strategies on tasks such as computing financial ratios and forecasting earnings before interest, taxes, depreciation and amortization (EBITDA) and sales. Interestingly, the most accurate results, aligning most closely with officially reported ratios and expert forecasts, are achieved with zero-shot prompting. This technique refers to providing no prior training examples before the model responds to a task. This suggests that in the context of financial information processing, zero-shot prompts may outperform more structured approaches, as models are required to reason independently.

However, the study has notable limitations. First, only three LLMs are included, all of which have considerably lower usage than widely adopted models such as ChatGPT, Gemini or Claude. Second, the results may have become outdated as newer models have been released since the study's conclusion. Third, the finding that zero-shot prompts outperform structured prompting contradicts general prompt engineering literature. Knoth et al. (2024) suggest that more structured prompting yields greater outputs. This is further supported by Wang et al. (2026), who find prompt engineering to benefit model performance, although in the context of healthcare. Meanwhile, literature on the use of AI in financial information processing suggests wider implications for PLCs.

Tieu and Tran (2026) highlight that ML (utilized by AI tools) integration within Vietnamese PLCs supports a dual purpose in compliance and fraud mitigation. ML improves the detection of abnormal patterns in financial reporting oversight, which strengthens compliance and reduces fraud risk. Support for AI's beneficial integration is shown by Ding et al. (2020), who compare ML predictions with management forecasts using data from U.S. insurance firms. Their results show that ML models outperform managerial estimates in 80 percent of cases. However, limited comparable studies and model differences prevent the findings from being more generalized.

Other developments include possibilities of decreasing auditing work during financial reporting by using AI-powered drones for capturing and analyzing footage for inventory inspections (Alhazmi et al., 2025). Additionally, Li (2026), in examining the effect of AI adoption on a firm's annual report disclosure quality, finds that AI improves financial reporting quality by strengthening internal information processing and reducing manual errors. However, this study considers the use of AI only through filed AI patent applications, which does not provide a quantifiable basis for overall AI adoption within firms. Yet, concerns around AI reliability exist, with practical consequences evident in the Deloitte case in 2025 (see [Appendix B](#)).

One of the most documented AI reliability issues is hallucination, which refers to LLMs convincingly generating misleading and fabricated information (Granstedt et al., 2025; Lee et al., 2025; Sood et al., 2025; Tirumala et al., 2024). Yet the literature on financial information processing has not examined how, or whether, optimizing the origin of online information for LLMs impacts the accuracy of AI-generated responses. As Cheng (2025) notes, some AI systems use automated web crawlers to extract online information, which is then merged into training datasets. These datasets are used to develop models that generate user responses. Connecting the issue of hallucinations with the use of web crawlers suggests that optimization strategies focused on the origin of online information may be beneficial for accuracy. One such strategy is answer engine optimization.

2.9 From Search Engine Optimization to AEO

Erdmann et al. (2022) define Search Engine Optimization (SEO) as a strategy for improving the visibility of a company's online content in search engine results. As consumers search for products, services or information through platforms such as Google, SEO focuses on website structure, keywords and content quality to support a high ranking in search results (Malaga, 2010; Veglis & Giomelakis, 2021, pp. 1-7). The intention is to increase user traffic to a specific website.

While SEO targets traditional search engines, the shift toward Answer Engine Optimization (AEO) concentrates on visibility in AI-generated responses. Contrary to traditional search ranking, AEO focuses on influencing the answer provided by an answer engine. The objective of a company utilizing AEO is to appear more frequently in top results when similar prompts are given by users (Chen et al., 2025). This emerging approach to visibility optimization is also occasionally referred to as generative engine optimization or artificial intelligence optimization.

2.9.1 Emergence of Answer Engine Optimization

Historically, SEO has served as the leading internet marketing tool for establishing a company's online presence. In the 1990s, early adopters benefited from increased visibility and monetization opportunities by appearing at the top of search engine results (Bezhovski, 2015). Initial ranking algorithms were primarily based on website meta-tags such as keywords and content descriptions (ibid.). Over time, algorithmic ranking has expanded to include a broader set of factors, alongside paid content placement via platform facilitators such as Google. Ranking factors include technical and user experience features such as page loading speed, mobile accessibility and the use of Hypertext Transfer Protocol Secure (HTTPS) to ensure encrypted and secure user connections (Almadhoun & Malim, 2025; Google, 2025a; Mavridis & Symeonidis, 2015).

The ongoing phenomenon of increased AI search at the expense of traditional search has shifted the focus of companies toward AEO. For instance, as of January 2026, ChatGPT reportedly has over 700 million weekly active users (OpenAI, 2026). Simultaneously, for queries where an AI Overview is present and the brand is not cited, Google's organic click-through rate declined by 65.2 percent year-on-year in 2025, as reported by Seer Interactive (McDonald, 2025). Although this rate is not an official disclosure and the sample size was limited to 42 client organizations with 3,199 queries, the results nonetheless indicate support for increased user adoption of AI.

As a result, companies seeking to remain visible and accurately represented are shifting from SEO to AEO as LLMs increasingly mediate information (Chen et al., 2025). However, academic literature on AEO is currently limited to a few preprint papers. Therefore, the following sections cover the four AEO areas relevant to this thesis and their potential impact on alignment of AI-generated responses with official PLC disclosures.

2.9.2 Foundational Strategies for Answer Engine Optimization

Current preprint studies argue that structured and accessible data minimizes the risk of hallucinations by LLMs (Alansari & Luqman, 2025; Rumiantsau et al., 2024). This aligns closely with AEO, which focuses on establishing website information in a structured, authoritative and citable form that AI systems can interpret and select when generating answers (Chen et al., 2025; Smajić et al., 2025). The foundational areas for optimizing website presence are arguably schema markup, content structure, technical SEO and E-E-A-T.

Schema markup is structured data added to a website to label and describe content. The vocabulary is published and maintained by Schema.org (2026). Schema.org was founded through a collaboration between Google, Microsoft, Yahoo and Yandex, with its schemas continuously updated through community effort (ibid.). Dekhtiar et al. (2018) define structured data as information that can be organized into rows and columns while maintaining all of its content. **Figure 1** provides an example of schema markup in use, based on Schema.org vocabulary. Information is clearly categorized in the data format of JavaScript Object Notation for Linked Data (JSON-LD). Nanjappan et al. (2017) note that JSON is easily generable by computer since it is built on a collection of name-value pairs and an ordered list of values. In theory, AI crawlers can identify and extract this information unambiguously.

Structured representation of the JSON-LD example.

@type	Organization
url	http://www.your-company-site.com
contactPoint	
@type	ContactPoint
telephone	+1-401-555-1212
contactType	customer service

Figure 1: Example of schema markup for customer service phone number from Schema.org (2026)

Cheng et al. (2025) find that individuals direct more questions to LLMs than to humans. This behavior highlights the value of structuring website content around question formats, such as “What is...”. Additionally, AEO practice commonly uses clear sections, short paragraphs and concise wording to support AI interpretation. This structural part of AEO covers only one side of the process, while technical SEO determines how effectively AI systems access the content. Other notable factors exist for technical SEO, such as mobile accessibility, which ensures webpages display correctly across mobile screen sizes (Schubert, 2016).

Kinnunen (2020) finds that the removal of redundant code increases webpage loading speed. This finding aligns with technical SEO, as reducing unnecessary code can improve loading speed by decreasing file size. Loading speed is also included in Google’s Core Web Vitals, which are used to evaluate webpage performance (Google, 2025a). Other notable factors exist for technical SEO, such as mobile accessibility, which ensures webpages display correctly across mobile screen sizes (Schubert, 2016).

Google states that their automated ranking systems are designed to prioritize reliable information for users (Google, 2025b). For this purpose, Google relies on the E-E-A-T framework, which includes experience, expertise, authoritativeness and trustworthiness (ibid.). **Table 1** summarizes how these qualities may be demonstrated from a company’s perspective. Consequently, AI systems similarly prioritize information that aligns with these reliability indicators (Chen et al., 2025).

E Experience	operation examples, case descriptions or demonstration of other activities
E Expertise	information about the author, credentials or qualifications, professional background
A Authoritativeness	“About us” page with company background, collaborations or partnerships with other organizations
T Trustworthiness	contact information, date stamps, HTTPS for secure connections and clear privacy disclosure

Table 1: Examples of E-E-A-T from a company perspective

2.9.3 Theoretical Benefits of Answer Engine Optimization

Consider a hypothetical publicly listed company, ExampleCorp, that notices AI-generated earnings summaries providing false information. While official reports are correct, AI occasionally mixes older quarterly earnings report figures or hallucinates completely.

To improve AI search results, ExampleCorp first adds schema markup to label sections of its IR webpage, such as reports. While schema markup structures webpage content, it does not extend to labeling specific numbers. Nevertheless, it supports AI systems in locating prompted information (Schema.org, 2026).

The company also organizes frequently asked information in question-and-answer sections, such as “What were Q4 revenues” and “When is the next quarterly earnings call? Such content structure allows AI to provide straightforward answers, citing ExampleCorp (Chen et al., 2025).

ExampleCorp simplifies paragraphs and wording to improve AI interpretation. Page loading speed and site performance in search engines are improved by removing unnecessary code and optimizing structure (Google, 2025a; Kinnunen, 2020).

Later, the company improves its credibility by adding management information, date stamps and securing its pages with HTTPS. These additions demonstrate E-E-A-T factors and may enhance trustworthiness for AI (Google, 2025b; Chen et al., 2025

Together, these practices show how a hypothetical firm can structure its online disclosures in the AI era. Structured, labeled and organized content may support AI in locating relevant information. Whether a relationship between AEO and the accuracy of AI-generated quarterly earnings summaries exists is the focus of this thesis.

2.10 Theoretical Framework and Hypotheses

To address the research question (*RQ: What is the relationship between a company’s Answer Engine Optimization and the accuracy of figures in AI-generated financial summaries?*), a combination of three theories is used to form a basis for this study. Given the absence of empirical studies on the topic, algorithmic opacity, signaling theory and information asymmetry are applied.

2.10.1 Relevant Theories

Algorithmic opacity refers to the lack of transparency or inability of AI in specifying why and how a decision is made to a user (Glikson & Woolley, 2020). Such opacity arises from the structure of AI, which consists of artificial neural networks that function as a mathematical tool with neurons similar to a human brain. These networks comprise an input, internal processing and a response as an output. The network contains multiple layers where the weights between neurons are continuously adjusted according to a learning rule to minimize the error between predicted and actual values (Goz et al., 2019; Jiménez-Pérez et al., 2023; Singh et al., 2019).

Figure 2 illustrates a simple neural network, where information flows forward in a single direction from the input through the hidden layer to the

output. The hidden layer represents the unknown internal processing that contributes to algorithmic opacity. For the mathematical equation of the output value, see [Appendix C](#).

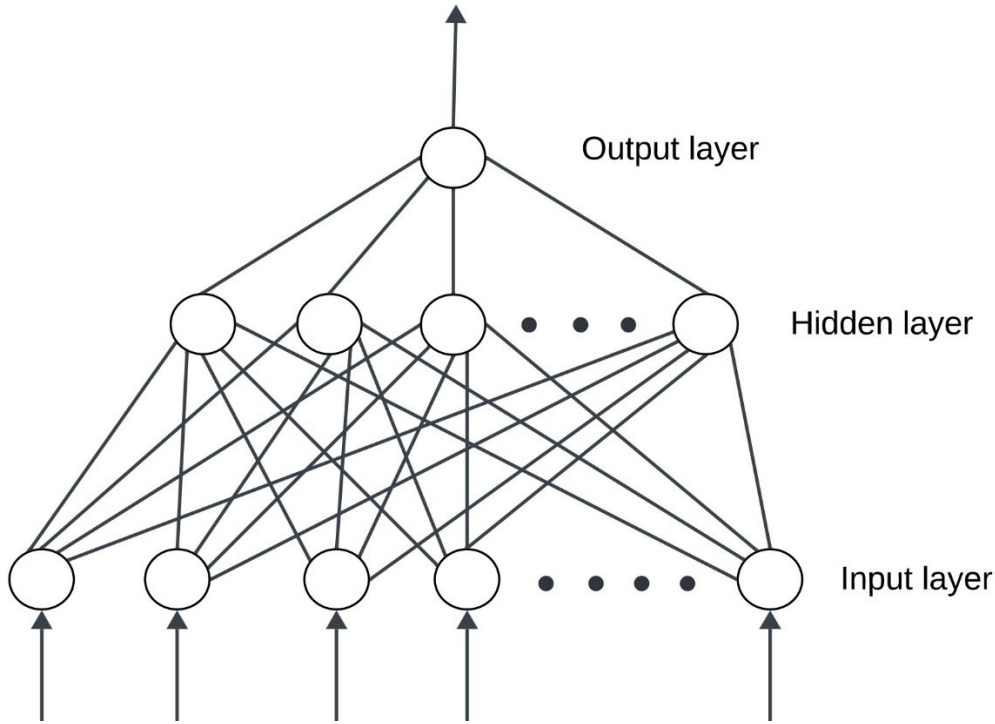


Figure 2: Three layered feed-forward neural network

Therefore, we arrive at the common depiction of AI systems as “black boxes” (Mühlbacher et al., 2014; Ribeiro et al., 2016; Ribeiro & Thill, 2022). Algorithms in neural networks require less preprocessing (e.g. filling information gaps) with structured data (Hollmann et al., 2025). Recent empirical results of Mohammed et al. (2025) find that inconsistent or less structured representation of information leads to higher deviation from ground truth values across 19 ML algorithms. However, these algorithms were not fine-tuned, meaning they were not trained for specific tasks. This contrasts how popular AI models are developed.

Algorithmic opacity suggests that AI can function unpredictably, especially when processing unstructured data. Such complexity may increase the risk of hallucinations in the information presented to users. Since AEO concentrates on optimizing a firm’s online disclosures through accessible, structured, citable and authoritative information, it theoretically supports mitigating hallucinations. As a result, it may influence the accuracy of AI-generated outputs presented to investors.

When including the perspectives of PLCs and investors under signaling theory (Connelly et al., 2011), information asymmetry is certain. PLCs possess

private information that is periodically published as the latest representation to investors. However, sharing this information requires costs through preparation and release of online disclosures, while possibly triggering market reactions. Nonetheless, such disclosures provide an incentive for investors to act on new information.

PLCs that optimize their online disclosures may reduce AI hallucinations and thereby lower information asymmetry. The release of financial disclosures additionally signals investors to obtain the latest information. In contrast, inaccurate AI-generated information may exacerbate the information imbalance, as the output fails to address the knowledge gap between investors and PLCs. **Figure 3** illustrates the theoretical framework developed as a basis for the hypotheses.

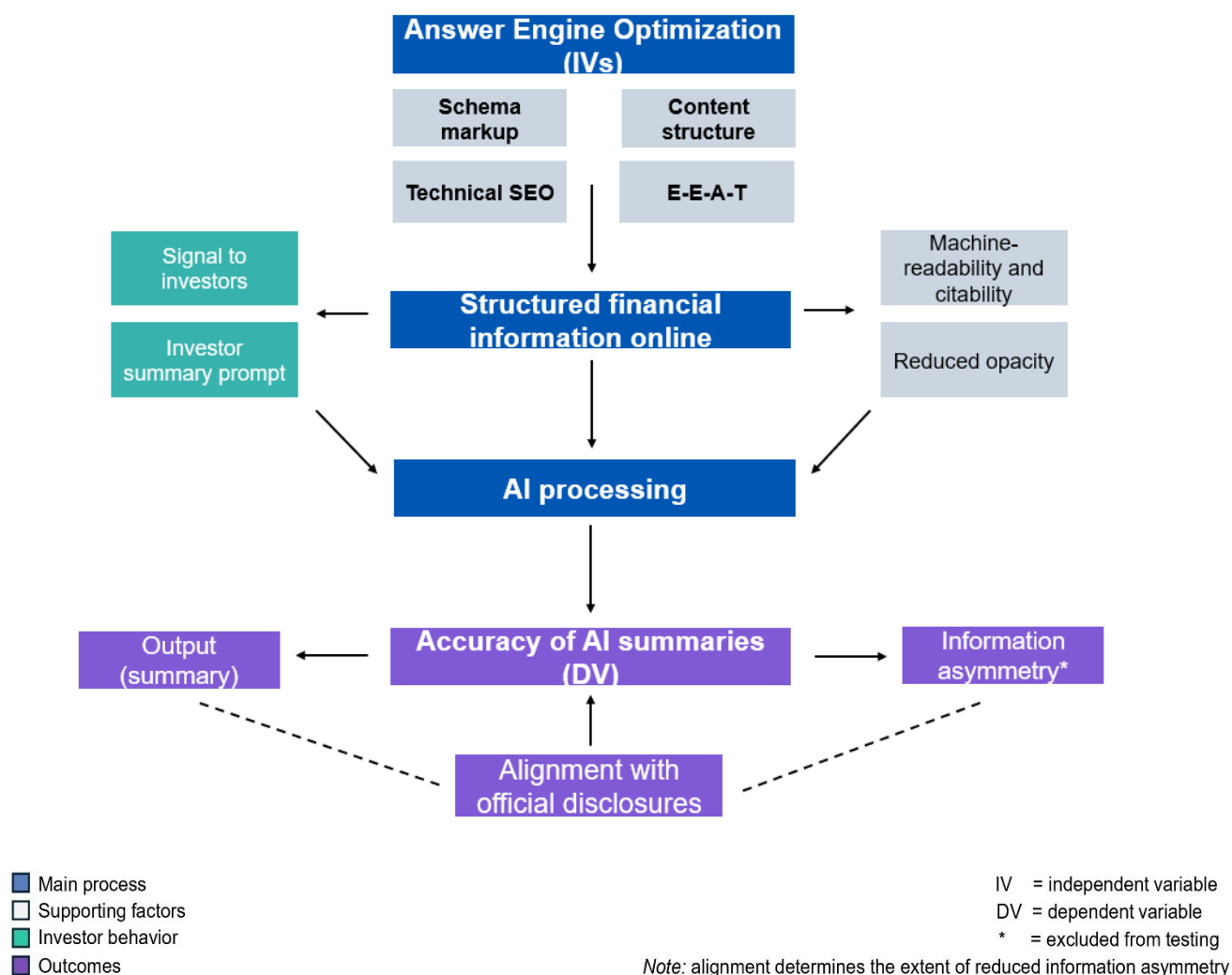


Figure 3: Theoretical framework

2.10.2 Hypotheses

Based on the theoretical framework in [Figure 3](#), the following hypotheses test the relationships between Answer Engine Optimization (AEO) and the accuracy of figures in AI-generated quarterly earnings summaries in terms of alignment with official disclosures. The primary hypothesis H1 proposes that a listed company's AEO score has a positive relationship with the accuracy of AI-generated financial summaries. To compare potential effects of timely information and citability, H2 suggests that the accuracy of AI-generated financial summaries differs between newly published and older quarterly reports. H3a-d explore the association of specific AEO areas, including schema markup, content structure, technical SEO and E-E-A-T, with the accuracy of AI-generated financial summaries. These hypotheses are summarized in [Table 2](#), providing a basis for a quantitative approach.

#	Hypothesis Statement	Expected Relationship
H1	A listed company's AEO score has a positive relationship with the accuracy of AI-generated financial summaries	Directional (positive)
H2	Accuracy of AI-generated summaries differs between newly published quarterly reports and older reports	Differential
H3a	Schema markup implementation is positively associated with accuracy	Directional (positive)
H3b	Content structure implementation is positively associated with accuracy	Directional (positive)
H3c	Technical SEO implementation is positively associated with accuracy	Directional (positive)
H3d	E-E-A-T implementation is positively associated with accuracy	Directional (positive)

Table 2: Summary of hypotheses

3 Methodology

3.1 Research Design

The nature of the research question (*RQ: What is the relationship between a company's Answer Engine Optimization and the accuracy of figures in AI-generated financial summaries?*) guided the research toward a quantitative approach. This design was supported by the use of an internal auditing tool from IDX Inc. that allowed for assessing the optimization of IR websites for answer engines. The dataset was constructed with both primary and secondary data, consisting of numerical values for statistical testing.

As PLCs listed on U.S. exchanges are mandated to file quarterly reports by the Securities and Exchange Commission (Office of the Federal Register, 2026), companies were selected at random from those publishing reports during the study period between January 12 and February 20, 2026. All companies in the sample are listed on either the New York Stock Exchange or Nasdaq. The final sample ($n = 105$) consists predominantly of U.S. -based companies, including several foreign multinationals with U.S. listings (see [Appendix D](#)).

3.2 Operationalization

To provide a proxy for a company's AEO without an in-depth analysis of each IR website, an internal auditing tool with 94% accuracy rate was utilized. After inserting a selected company's IR website URL, the tool automatically scores the digital presence on a scale of 0 to 100 based on independent variables that a firm can control. These variables include the overall score, schema markup, content structure, technical SEO and E-E-A-T (Experience, Expertise, Authoritativeness and Trustworthiness). The specific scores represent weighted averages of various factors within each variable, while the overall AEO score is calculated based on weightings of 30% for schema markup, 25% for content structure, 25% for technical SEO and 20% for E-E-A-T.

Regarding the secondary data, two of the latest quarterly reports for each selected company were obtained to provide a basis for comparison, given that AI outputs may deviate from officially disclosed figures. This allowed the generation of financial report summaries for both the current and previous quarterly periods. The official figures provided the known true values to evaluate the AI output. Each reported figure was scored, where one represented full alignment with official disclosures and zero represented non-alignment.

To generate data for the dependent variable of AI accuracy, ChatGPT-5.2 was prompted to provide figures for each selected company's earnings. This model was chosen due to its popularity in terms of monthly users. It was the latest GPT model at the time of data collection, making it representative for the purposes of this thesis.

Data was collected using an anonymous browser with new chat logs for each generation. This ensured each prompt remained isolated, preventing the model from mimicking previous responses inside chat logs. The prompt was standardized to reflect how a retail investor might inquire about a company's financials. The broad term "key financials" was used to allow the model flexibility in identifying relevant figures, since key financial figures tend to be labeled in the reported highlights. After testing various prompts and assessing their typical output, a final prompt was used for both pre- and post-earnings reports to direct the AI toward the IR websites:

"Summarize [Company X's] latest quarterly earnings report. Focus on the key financial figures and avoid commentary or external information."

Accuracy was determined by dividing the number of accurate figures by the total number of figures provided in the output. This was calculated for both pre- and post-earnings summaries to arrive at two distinct accuracy percentages. An average of these was then calculated to determine a proxy for overall accuracy for each selected company.

3.3 Data Collection and Cleaning

The data was collected between January 12 and February 20, 2026. The gathering stage required a precise schedule, as summaries had to be generated without delays following a report's publication. For the selected companies, fourth quarter summaries were generated within 24 hours of release, with some summaries generated as early as 30 minutes after publication. This schedule was intended to reflect investor behavior. It also increased the likelihood of the AI retrieving figures directly from official IR websites over third-party media sources.

However, the third quarter results were generated approximately 24 to 48 hours before the fourth quarter reports were published. After generating the data, segment-specific figures, such as regional breakdowns, were excluded. This reduced the raw dataset to 92 unique figure categories across 210 summaries. A figure was counted as accurate if its rounding matched the officially reported value.

3.4 Data Analysis

After data was collected and cleaned, IBM SPSS Statistics software was used for analysis, as it allowed for rapid visual output. The first statistical test was a linear correlation to assess H1 with the formula:

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n \sum x^2 - (\sum x)^2] [n \sum y^2 - (\sum y)^2]}}$$

Since the distribution of accuracy scores appeared not to follow a normal distribution, Spearman's correlation was included as a nonparametric alternative. As such, no assumptions about the distribution are made, with the formula:

$$r_s = 1 - \frac{6 \sum D^2}{n(n^2 - 1)}$$

The correlations were intended to assess whether a statistically significant relationship exists between overall AEO score and average accuracy (H1). Subsequently, a paired samples t-test was conducted to assess whether AI accuracy differed between older and newly published quarterly reports (H2). The following formula was used for the t-test:

$$t = \frac{\bar{d}}{\frac{s_d}{\sqrt{n}}}$$

To identify which AEO area is most associated with AI accuracy, a linear multiple regression was conducted (H3). Without definitive evidence of the distribution's nature, the Central Limit Theorem justifies the assumption of approximate normality for a sample size exceeding 30 observations (Kwak & Kim, 2017). This supports the use of parametric tests, including the linear multiple regression, where the formula is:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4 + \varepsilon$$

4 Findings

The sample of 105 companies reveals a mean overall AEO score of 50.40 ($Mdn = 55, SD = 10.76$). Accuracy scores remain above 94% in both pre- and post-publication, although post-accuracy shows higher variation ($SD = 17.98\%$ vs. 7.44%). Among the AEO areas, technical SEO yields the highest mean score (77.90), while schema markup shows the lowest (10.76). The majority of companies score zero in schema markup ($Mdn = 0$). Full descriptive statistics are presented in [Figure 4](#) below.

	Mean	Median	Standard Deviation	Max	Min
Overall AEO	50,40	55	10,76	73	19
Pre-Acc. (%)	96,76	100	7,44	100	67
Post-Acc. (%)	94,66	100	17,98	100	0

Schema markup	10,76	0	22,71	80	0
Content Structure	57,86	75	23,59	100	0
Technical SEO	77,90	80	23,34	100	50
E-E-A-T	42,38	35	22,90	80	0

Figure 4: Descriptive statistics

4.1 Pearson and Spearman Correlations

The correlations were first run between the overall AEO score and mean accuracy. The Pearson correlation reveals a statistically significant positive relationship between the variables ($r = .376, p < .001$). However, the correlation coefficient is weak.

Correlations			
		Overall_Score	Accuracy_AVG
Overall_Score	Pearson Correlation	1	.376***
	Sig. (2-tailed)		<.001
	N	105	105
Accuracy_AVG	Pearson Correlation	.376***	1
	Sig. (2-tailed)	<.001	
	N	105	105

***. Correlation at 0.001 (2-tailed)

Figure 5: Snippet of Pearson correlation

As the distribution of average accuracy scores appeared heavily skewed, with approximately 77% of companies scoring between 93% and 100%, Spearman

correlation was run as an alternative that does not assume normality. The results similarly indicate a statistically significant positive relationship ($r = .400, p < .001$), with a slightly stronger, moderate correlation coefficient.

Correlations				
			Overall_Score	Accuracy_AVG
Spearman's rho	Overall_Score	Correlation Coefficient	1.000	.400***
		Sig. (2-tailed)	.	<.001
		N	105	105
	Accuracy_AVG	Correlation Coefficient	.400***	1.000
		Sig. (2-tailed)	<.001	.
		N	105	105

***. Correlation is significant at the 0.001 level (2-tailed).

Figure 6: Snippet of Spearman's rho correlations

The following scatter plot illustrates the distribution of data points across all companies. The x-axis represents the overall AEO score, while the y-axis represents the average accuracy percentage. The model shows an insignificant explanatory power ($R^2 = .141$), which is consistent with the weak to moderate correlation coefficients.

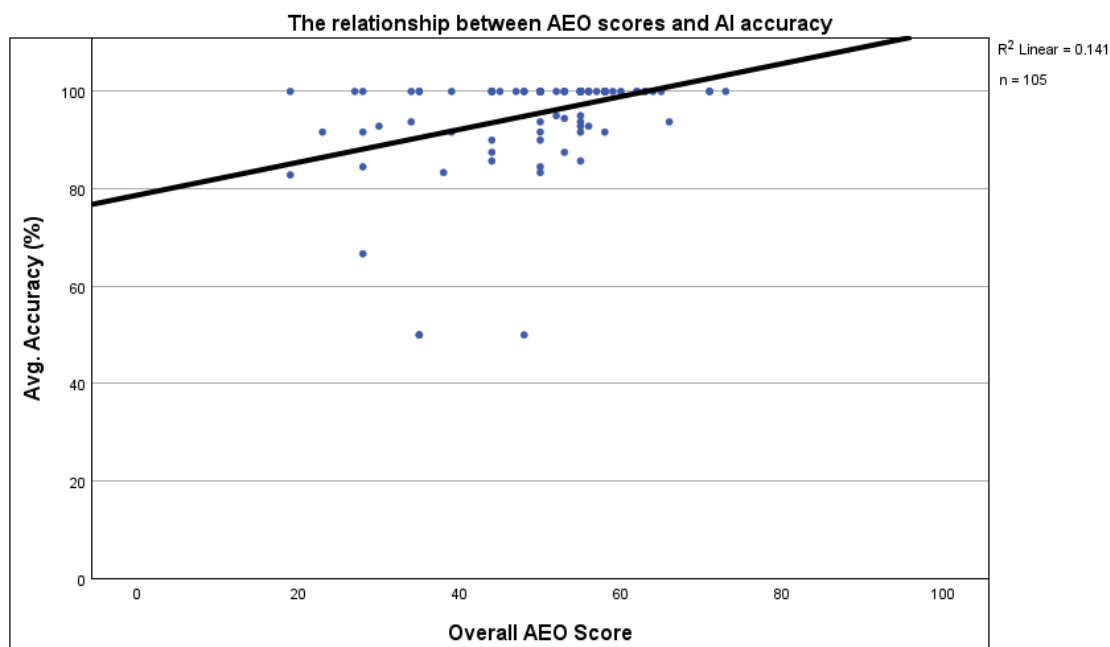


Figure 7: Linear relationship chart between overall AEO score and average accuracy

As for the relationships between AEO areas and accuracy, the Pearson correlation table reveals that content structure ($r = .326$) and technical SEO ($r = .255$) show the strongest positive correlations with accuracy, although both remain weak. Schema markup ($r = .093$) and E-E-A-T ($r = -.122$) show insignificant relationships. Notably, technical SEO and E-E-A-T demonstrate a strong negative correlation ($r = -.771$). Spearman's rho correlations show similar results and can be accessed from [Appendix E](#).

		Correlations					
		Overall_Score	Technical_SEO	Content_Structure	Schema_Markup	EEAT	Accuracy_AVG
Overall_Score	Pearson Correlation	1	.342***	.858***	.313**	.067	.376***
	Sig. (2-tailed)		<.001	<.001	.001	.494	<.001
	N	105	105	105	105	105	105
Technical_SEO	Pearson Correlation	.342***	1	.436***	-.472***	-.771***	.255**
	Sig. (2-tailed)	<.001		<.001	<.001	<.001	.009
	N	105	105	105	105	105	105
Content_Structure	Pearson Correlation	.858***	.436***	1	-.103	-.217*	.326***
	Sig. (2-tailed)	<.001	<.001		.295	.026	<.001
	N	105	105	105	105	105	105
Schema_Markup	Pearson Correlation	.313**	-.472***	-.103	1	.521***	.093
	Sig. (2-tailed)	.001	<.001	.295		<.001	.345
	N	105	105	105	105	105	105
EEAT	Pearson Correlation	.067	-.771***	-.217*	.521***	1	-.122
	Sig. (2-tailed)	.494	<.001	.026	<.001		.217
	N	105	105	105	105	105	105
Accuracy_AVG	Pearson Correlation	.376***	.255**	.326***	.093	-.122	1
	Sig. (2-tailed)	<.001	.009	<.001	.345	.217	
	N	105	105	105	105	105	105

***. Correlation at 0.001 (2-tailed)
 **. Correlation is significant at the 0.01 level (2-tailed).
 *. Correlation is significant at the 0.05 level (2-tailed).

Figure 8: Pearson Correlations Table

4.2 Paired Samples T-test

Based on the results, pre- and post-accuracy scores are not correlated ($r = -.016$, $p = .872$). The paired samples t-test shows no statistically significant difference in mean accuracy ($t(104) = 1.098$, $p = .275$), as the p-value exceeds the .05 significance threshold.

Paired Samples Correlations				
	N	Correlation	Significance	
			One-Sided p	Two-Sided p
Pair 1 Accuracy_PRE & Accuracy_POST	105	-.016	.436	.872

Paired Samples Test									
		Paired Differences			95% Confidence Interval of the Difference		Significance		
		Mean	Std. Deviation	Std. Error Mean	Lower	Upper	t	df	
Pair 1	Accuracy_PRE - Accuracy_POST	2.0959889210	19.568932237	1.9097322398	-1.691081776	5.8830596180	1.098	104	
									One-Sided p Two-Sided p
									.137 .275

Figure 9: Pre- and post-accuracy t-test

4.3 Multiple Regression

The multiple regression model for AEO areas predicting accuracy is statistically significant ($p < .001$), with an adjusted R^2 of .134, explaining 13.4% of the variation in accuracy. Among the four predictors, schema markup ($B = .102$, $p = .029$) and content structure ($B = .094$, $p = .029$) are statistically significant, while technical SEO and E-E-A-T remain insignificant. When excluding E-E-A-T, the model improves slightly ($R_a^2 = 0.143$), explaining 14.3% of the variation in accuracy.

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	R Square Change	Change Statistics			
						F Change	df1	df2	Sig. F Change
1	.409 ^a	.168	.134	9.0021090065	.168	5.035	4	100	<.001

a. Predictors: (Constant), EEAT, Content_Structure, Schema_Markup, Technical_SEO

Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.	95.0% Confidence Interval for B	
		B	Std. Error	Beta			Lower Bound	Upper Bound
1	(Constant)	79.383	6.883		11.534	<.001	65.728	93.038
	Schema_Markup	.102	.046	.239	2.215	.029	.011	.193
	Content_Structure	.094	.043	.230	2.215	.029	.010	.179
	Technical_SEO	.119	.067	.288	1.791	.076	-.013	.252
	EEAT	.011	.064	.026	.171	.864	-.116	.138

a. Dependent Variable: Accuracy_AVG

Figure 10: Multiple regression model with AEO predictors

4.4 Summary of Findings

The table below summarizes the key empirical findings from the quantitative analysis. The results are indicative, providing a basis for discussion rather than conclusions. Similarly, it provides as a foundation for the following section, which covers the findings in the context of the study objectives.

Metric / Analysis	Statistical Result	Finding
Median for Schema markup	$Mdn = 0$	The majority of the sampled companies score zero in schema markup
Standard Deviation (Accuracy)	$SD_{pre} = 7.44$ $SD_{post} = 17.98$	Post-accuracy shows higher variation than pre-accuracy
Pearson & Spearman Correlations	$r = .376, r_s = .400,$ ($p < .001$)	A slight, statistically significant positive correlation exists between overall AEO score and accuracy
Technical SEO vs. E-E-A-T	$r = -.771$	A strong negative correlation exists between the two variables.
Paired Samples T-test	$t(104) = 1.098, p = .275$	No statistically significant difference in accuracy before and after publication
Multiple Regression Models	$R_a^2 = .134$ & $.143$ $F = 5.04$ & 6.78 ($p < .001$)	AEO efforts explain 13.4% to 14.3% of the variation in accuracy

Table 3: Summary of findings

5 Discussion and Analysis

5.1 Interpretation of Key Findings

Starting with the descriptive statistics, the median schema markup score of zero suggests that structured data for answer engines remains largely unexplored among PLCs. In fact, only around 21% of sampled PLCs had implemented schemas to some extent. However, it is possible that other data vocabularies were implemented but not captured by the auditing tool. Nevertheless, structured data for answer engines appears to be emerging within standard IR practices.

As for the accuracy variation between Acc_{PRE} and Acc_{POST} , a notable factor is that 3 out of 105 sampled companies received completely inaccurate AI output. This was likely due to hallucination or confusion with the reporting format of quarterly earnings, when compared against the officially reported figures. Consequently, SD_{post} was considerably impacted, as when companies with zero accuracy are excluded, SD_{post} reduces to 7.68. This is within a comparable range to SD_{pre} of 7.44, suggesting that the documented instances were likely due to random error rather than systematic inaccuracies.

5.1.1 Correlations

The correlations confirmed a statistically significant relationship between AEO efforts and accuracy. However, the weak to moderate coefficients indicate a partial relationship, suggesting that additional variables may contribute to explaining accuracy. Furthermore, the slightly higher Spearman correlation raises the question of whether the relationship between AEO and accuracy is linear or non-linear in nature.

Figure 7 illustrates that companies with low AEO scores can still achieve full accuracy. However, such instances appear less frequent at lower scores. The high average accuracy scores suggest a ceiling effect, whereby the benefit of further optimization may diminish beyond a certain level.

The findings may also be largely due to the distribution of AEO scores among sampled PLCs. The largest portion of companies fall within the range of 52 to 59, with the second largest portion ranging from 44 to 51. Consequently, both tails, scores below 30 and above 70, are underrepresented. This may contribute to the consistency of accuracy results, while reducing the exploratory power of the correlations to some extent. **Figure 11** illustrates the distribution of overall AEO scores.

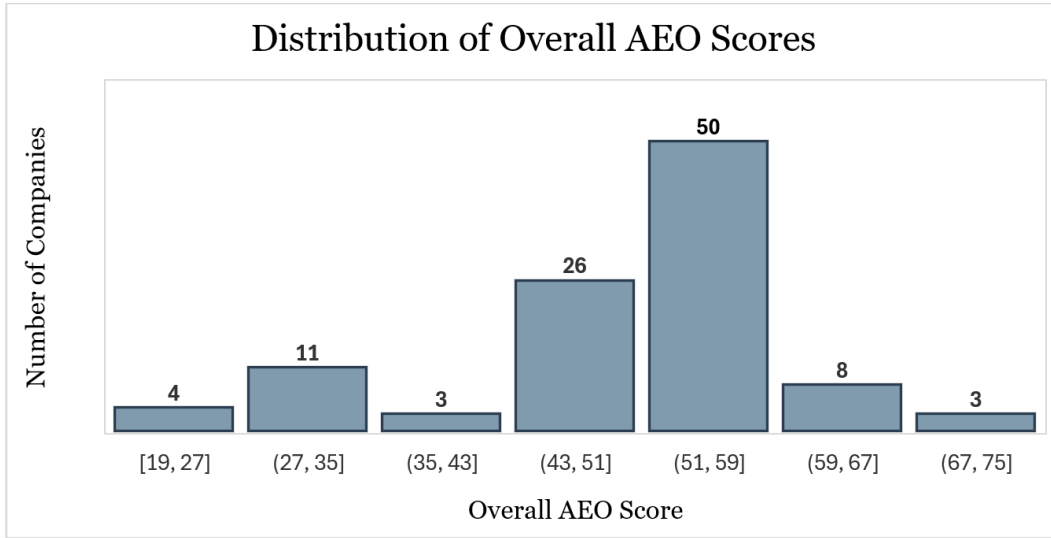


Figure 11: Distribution of overall AEO scores

Additionally, a strong negative correlation between technical SEO and E-E-A-T suggests that companies currently prioritize one AEO area at the expense of the other. This may reflect a trade-off between technical optimization and E-E-A-T priorities.

5.1.2 T-test and Multiple Regression Models

The t-test shows the variation between Acc_{pre} and Acc_{post} to be insignificant. This supports the assumption that answer engines are consistent in providing accurate information regardless of whether the quarterly report has been recently published or not.

The regression results using AEO areas as predictors explain between 13.4% and 14.3% of the variation in accuracy. The low R_a^2 suggests that additional factors beyond current predictors may contribute to explaining accuracy. Alternatively, the extent to which companies can influence AI accuracy may be limited. Nevertheless, the current findings should be replicated with a larger and more diverse dataset to reach more definitive conclusions. As the first results of this relationship, they represent an early indication of AEO's impact on accuracy.

5.2 Outcomes for the Hypotheses

H1 proposed that a listed company's AEO score has a positive relationship with the accuracy of AI-generated financial summaries. The results from the correlations support the hypothesis that a statistically significant relationship exists ($p < .001$). However, the nature of this relationship, whether linear or non-linear, remains uncertain. This is mostly due to some companies with low AEO scores still achieving full accuracy.

H2 proposed that the accuracy of AI-generated summaries differs between newly published and older quarterly reports. This hypothesis is rejected based on the t-test results ($t(104) = 1.098, p = .275$) and further supported by comparable standard deviations when outliers are excluded. Therefore, the accuracy of AI-generated summaries appears consistent regardless of publication timing.

H3a proposed that schema markup implementation is positively associated with accuracy. This hypothesis is accepted, as the regression coefficient is statistically significant ($B = .102, p = .029$).

H3b proposed that content structure implementation is positively associated with accuracy. This hypothesis is similarly accepted, as the regression coefficient is statistically significant ($B = .094, p = .029$).

H3c proposed that technical SEO implementation is positively associated with accuracy. This hypothesis is rejected, as the p -value exceeds the threshold of .05. Essentially, it appears insignificant ($p = .076$).

H3d proposed that E-E-A-T implementation is positively associated with accuracy. This hypothesis is rejected, as the p -value exceeds the threshold of .05. The value appears highly insignificant ($p = .864$).

Overall H3a-d proposed that specific AEO areas are positively associated with accuracy, where an increase in a predictor would increase accuracy and vice versa. Technical SEO (H3c) and E-E-A-T (H3d) appear insignificant when included in the same model, although this is likely due to the strong negative correlation between the two variables. However, this does not exclude the possibility that with a larger sample size, the significance of these variables may change.

5.3 Outcome for the Research Question

What is the relationship between a company's Answer Engine Optimization and the accuracy of figures in AI-generated financial summaries? Based on the results, a statistically significant yet moderate relationship exists between AEO and accuracy. Among the AEO areas, schema markup and content structure are found to be significant predictors, while technical SEO and E-E-A-T remain insignificant.

Figure 3 provided the theoretical basis for the study. The moderate relationship aligns with the understanding that AI models generate responses through complex layers, which creates algorithmic opacity. Because AEO areas explain a fraction of the variation in accuracy (13.4% to 14.3%), the results suggest that a model's internal processes remain a notable predictor. Furthermore, **Figure 7** illustrates that high AEO scores do not guarantee full accuracy and low scores do not necessarily cause inaccuracy. This supports the unpredictability of AI as a "black box". The complexity of AI model processes therefore appears to remain a fundamental limitation to AEO's effect on accuracy improvement. Currently, it appears that there is no direct control over the output accuracy generated by AI, but influencing is possible.

As for investors acting on quarterly reports, no timing advantage was found, with accuracy remaining consistent regardless of publication timing. AI intermediaries appear to be well adapted to investor behavior (use of such tools for summarization) after reports are published. Theoretically, this may quicken the price adjustments on secondary markets if decisions are based solely on AI output and not manually verified. Additionally, the significance of schema markup and content structure supports that structured and machine-readable disclosures can reduce the information asymmetry that AI hallucinations would otherwise worsen.

Overall, the findings indicate that optimization (AEO) can meaningfully improve the alignment of AI-generated financial information. Yet, the influence appears limited, with additional factors likely contributing beyond current predictors.

5.4 Limitations

Due to time constraints, the sample size is intended to provide an early indication. However, a larger sample would yield more reliable results. The limited sample size likely results in the lower explanatory power (R^2). Moreover, the narrow AEO score range with underrepresented extremes may have

weakened the correlations. This was due to quarterly report publications being outside the control during data collection.

Since data collection, OpenAI has published a newer model, which may not yield identical results to the model used in this study. The results are also dependent on the specific prompt used during data collection. Additionally, accuracy percentages were calculated based on AI output, which automatically penalizes companies with fewer reported figures.

A further limitation is that the relationship between AEO and accuracy may not necessarily be linear. Should the relationship follow a different distribution, the current test results may not provide valid indications for AI accuracy.

Finally, all AEO scores were automatically audited, leaving a slight possibility of false results given the tool's reported accuracy of 94%. However, this approach was the most viable option given the time constraints and individual work requirements.

6 Conclusions

This thesis aimed to provide an early indication of the relationship between answer engine optimization and the accuracy of AI-generated financial summaries. A quantitative approach was utilized, finding a statistically significant yet moderate relationship between the variables. The results suggest a ceiling effect, where the high accuracy scores across sampled companies limit the impact of AEO on AI accuracy. However, the results likely underestimate the true strength of this relationship due to the limited sample size and high frequency of zero values for schema markup.

6.1 Main Findings

Three hypotheses were tested alongside the research question examining the relationship between AEO and the accuracy of figures in AI-generated financial summaries. The results showed a statistically significant ($p < .001$) yet moderate relationship between the variables, with AEO areas explaining between 13.4% and 14.3% of accuracy variation in regression models (R_a^2). The moderate relationship and limited influence align with the understanding behind algorithmic opacity, where the complexity of AI model processes may lead to inconsistencies in outputs.

H1 is accepted, as both Pearson and Spearman correlations confirmed a significant positive relationship. Whether the relationship is linear or non-linear, however, remains undetermined. H2 is rejected, as the paired samples t-test found no significant difference between Acc_{pre} and Acc_{post} . This suggests AI accuracy remains consistent regardless of timing. For H3a-d, schema markup and content structure are accepted, while technical SEO and E-E-A-T are rejected. The rejections are likely due to a strong negative correlation between technical SEO and E-E-A-T scores.

#	Hypothesis Statement	Expected Relationship	Result
H1	A listed company's AEO score has a positive relationship with the accuracy of AI-generated financial summaries	Directional (positive)	Accepted
H2	Accuracy of AI-generated summaries differs between newly published quarterly reports and older reports	Differential	Rejected

H3a	Schema markup implementation is positively associated with accuracy	Directional (positive)	Accepted
H3b	Content structure implementation is positively associated with accuracy	Directional (positive)	Accepted
H3c	Technical SEO implementation is positively associated with accuracy	Directional (positive)	Rejected
H3d	E-E-A-T implementation is positively associated with accuracy	Directional (positive)	Rejected

Table 4: Results from Hypotheses

Approximately 79% of sampled companies had no schema implementation, suggesting structured data remains an emerging rather than established IR practice. Together, AEO appears to influence AI accuracy to a limited extent, with the “black box” structure of AI models remaining a limitation regardless of optimization. The consistency of accuracy across publication timing appears beneficial for the growing investor use of AI tools for summarizing quarterly disclosures. Overall, AEO presents a meaningful way for PLCs to mitigate disruptions in the information flow that investors rely and act upon, even if no direct control over AI output accuracy is possible.

6.2 Implications for International Business

This thesis provided an early indication of AEO’s role in influencing the accuracy of AI-generated quarterly report summaries. With the rapid development of AI models, the representation of PLCs appears to be increasingly shifting toward AI intermediaries. These systems interpret, extract and present disclosures, including financial information, to investors. AI is therefore becoming an additional, yet indirect, communication channel between companies and investors. Consequently, PLCs should manage how their disclosures are interpreted and presented by AI output.

For PLCs aiming to influence their AI representation, AEO can be regarded as a new signaling cost of investor communication. Similar to how firms invest in preparing and publishing financial disclosures, optimizing digital presence for AI appears relevant for maintaining transparent investor relations. This is done by decreasing information asymmetry through accurate AI representation. Broadly, the transparency and visibility from AEO extends to all company disclosures, such as management strategy and sustainability. The results suggest that AEO efforts are positively associated with accuracy in AI-generated financial summaries. Without optimization, PLCs risk increasing investor reliance on information from unverified third-party

sources, such as Reddit, while the visibility of official disclosures may be compromised. Such instances occurred occasionally in AI output during data collection, where the primary source was a Reddit post.

For service providers facilitating AEO, the results suggest that optimization efforts are not random but meaningfully linked to accuracy. The findings provide an indicative validation that PLCs can be supported to improve and maintain accurate AI representation. Among the AEO areas, schema markup and content structure appear to yield the greatest benefit for accuracy. However, service providers should be aware of the potential trade-off between technical SEO and E-E-A-T, as improvements in one area may come at the expense of the other. This highlights the necessity of creating a balance between the areas adapted to each PLC's needs.

For monitoring purposes following IR website optimization, an accuracy indicator for financial figures using both Acc_{pre} and Acc_{post} may prove useful. This could track alignment against officially reported figures using a standardized prompt. In doing so, it would provide a proxy for hallucination rate. For more reliable results, multiple prompts should be included while ensuring information originates from the IR website itself.

6.3 Suggestions for Further Research

A natural extension would be to group a comparable sample of PLCs based on overall AEO scores (e.g. low = 0-35, medium = 36-64, high = 65-100) and replicate the statistical tests from this thesis alongside group-specific analysis. This would allow for a direct comparison with the current findings while ensuring sufficient representation across the score scale. To provide a suitable sample size estimation for a statistically acceptable conclusion on AEO influence, Slovin formula for sample size with 5% error margin is used:

$$n = \frac{N}{1 + N \times e^2}$$

According to the World Federation of Exchanges (2026) latest approximation for the number of PLCs globally is 49,054. Given the total population ($N = 49,054$) an estimated sample size would be 396.76 with rounded to 397. Therefore, the ideal and comfortable sample size would be over 400.

Future research should aim for a more balanced representation of companies with and without schema markup to obtain a more reliable estimate of the correlation with accuracy. Alternatively, non GPT-based model results would be interesting to compare against the current findings. This design remains open for any future research in this area.

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Appendices

Appendix A

The screenshot displays an iXBRL viewer interface. The main area shows a financial statement titled "Statement of Comprehensive Income for the Period from January 1 to December 31, 2017". The table compares data for "Jan. to Dec. 2017" and "Jan. to Dec. 2016". The "Fact Properties" sidebar on the right provides details for the selected fact: (ifrs) Licence fee income, with a value of US \$ 7,786,002 for the period 1 Jan 2017 to 31 Dec 2017. The sidebar also shows dimensions like Date, Fact Value, Accuracy, Change, Entity (LEI: 506700GE1G29325QX363), and Concept (ifrs:LicenceFeeIncome).

	Notes	Jan. to Dec. 2017	Jan. to Dec. 2016
		US\$	US\$
Fee revenue	3.1	7,786,002	6,245,640
Wages and salaries		-3,404,031	-2,700,694
Social contributions and expenses for pensions and care		-456,337	-376,780
Personnel expenses	3.2	-3,860,368	-3,077,474
Other operating expenses	3.3	-4,058,493	-3,348,252
Other operating income	3.4	539,527	201,019
Amortisation and depreciation expense	4.5/4.6	-214,955	-165,030
Operating surplus		191,713	-144,097
Subsidies and donations	3.5	61,454	158,354
Financial income/expense	3.6	-300	479
Net surplus		252,867	14,736

Screenshot example of inline eXtensible Business Reporting Language (iXBRL). For individual exploration, the source can be accessed at the link below.

Global Legal Entity Identifier Foundation (2017) “Annual Report 2017 (iXBRL viewer).” *LEI*. [Online]. Available from: <https://www.xbrl.org/ixbrl-samples/gleif-annual-report-2017/gleif-annual-report-2017-viewer.html> [Accessed on 14 February 2026].

Appendix B

Consequences of Inconsistent AI Outputs: Deloitte

In October 2025, Deloitte admitted to using AI to help prepare a government report for the Albanese administration in Australia. Due to hallucinations, this led to a repayment of \$440,000 AUD to the government, along with reputational harm. For more information, see the source below.

Dhanji, K. (2025) “Deloitte to pay money back to Albanese government after using AI in \$440,000 report.” *The Guardian*, 6 October, 08:46 CEST. [Online]. Available from:

<https://www.theguardian.com/australia-news/2025/oct/06/deloitte-to-pay-money-back-to-albanese-government-after-using-ai-in-440000-report>
[Accessed on 14 February 2026].

Appendix C

For two consecutive layers $(l - 1, l)$, the output value α_j is determined by the following equation:

$$\alpha_j = \Theta_j \left(b_j + \sum_{i=1}^m W_{i,j} \beta_i \right)$$

α_j : Output value of neuron j in layer l

β_i : Input value from the previous layer

Θ_j : Activation function

b_j : Threshold coefficient (bias)

$W_{i,j}$: Connection weight between neuron i and neuron j

m : Number of neurons in the previous layer $(l - 1)$

i : Position of neuron in the previous layer (1 to m)

This widely accepted formula demonstrates how raw data is transformed through weights mathematically within neural networks to generate a response.

Appendix D

List of Public Listed Companies (PLCs) in the Study Dataset

Company Name	Trading Symbol	Exchange
3M Company	MMM	NYSE
Advanced Micro Devices, Inc.	AMD	NASDAQ
Airbnb, Inc.	ABNB	NASDAQ
Allegion plc	ALLE	NYSE
Amazon.com, Inc.	AMZN	NASDAQ
American Express Company	AXP	NYSE
AngloGold Ashanti plc	AU	NYSE

Apollo Global Management, Inc.	APO	NYSE
Avery Dennison Corporation	AVY	NYSE
Bank of America Corporation	BAC	NYSE
Bank7 Corp.	BSVN	NASDAQ
BlackRock, Inc.	BLK	NYSE
The Boeing Company	BA	NYSE
Booking Holdings Inc.	BKNG	NASDAQ
Builders FirstSource, Inc.	BLDR	NYSE
Cadence Design Systems, Inc.	CDNS	NASDAQ
Carvana Co.	CVNA	NYSE
Caterpillar Inc.	CAT	NYSE
CEVA, Inc.	CEVA	NASDAQ
Chevron Corporation	CVX	NYSE
Citigroup Inc.	C	NYSE
Cloudflare, Inc.	NET	NYSE
CME Group Inc.	CME	NASDAQ
CNH Industrial N.V.	CNH	NYSE
Concrete Pumping Holdings, Inc.	BBCP	NASDAQ
CSX Corporation	CSX	NASDAQ
Danaher Corporation	DHR	NYSE
Deere & Company	DE	NYSE
Delta Air Lines, Inc.	DAL	NYSE
Devon Energy Corporation	DVN	NYSE
DTE Energy Company	DTE	NYSE
Dynatrace, Inc.	DT	NYSE
EQT Corporation	EQT	NYSE
Equinor ASA	EQNR	NYSE
Expro Group Holdings N.V.	XPRO	NYSE
Fastenal Company	FAST	NASDAQ
FirstEnergy Corp.	FE	NYSE
Fluor Corporation	FLR	NYSE
Ford Motor Company	F	NYSE
Garmin Ltd.	GRMN	NYSE
GE Vernova Inc.	GEV	NYSE
Genuine Parts Company	GPC	NYSE
Global Payments Inc.	GPN	NYSE
The Goldman Sachs Group, Inc.	GS	NYSE
Hecla Mining Company	HL	NYSE

Herc Holdings Inc.	HRI	NYSE
Hexcel Corporation	HXL	NYSE
Home BancShares, Inc.	HOMB	NYSE
Hudbay Minerals	HBM	NYSE
Insteel Industries Inc.	IIN	NYSE
Intel Corporation	INTC	NASDAQ
J.B. Hunt Transport Services, Inc.	JBHT	NASDAQ
Johnson & Johnson	JNJ	NYSE
JPMorgan Chase & Co.	JPM	NYSE
Kenvue Inc.	KVUE	NYSE
Kinder Morgan, Inc.	KMI	NYSE
Kinross Gold Corporation	KGC	NYSE
Live Nation Entertainment, Inc.	LYV	NYSE
McDonald's Corporation	MCD	NYSE
Medtronic plc	MDT	NYSE
Merck & Co., Inc.	MRK	NYSE
Meta Platforms, Inc.	META	NASDAQ
Microsoft Corporation	MSFT	NASDAQ
Moody's Corporation	MCO	NYSE
Morgan Stanley	MS	NYSE
Navient Corporation	NAVI	NASDAQ
Netflix, Inc.	NFLX	NASDAQ
Newmont Corporation	NEM	NYSE
Novo Nordisk A/S	NVO	NYSE
Occidental Petroleum Corporation	OXY	NYSE
Otis Worldwide Corporation	OTIS	NYSE
Palantir Technologies Inc.	PLTR	NYSE
Pan American Silver Corp.	PAAS	NYSE
PayPal Holdings, Inc.	PYPL	NASDAQ
PepsiCo, Inc.	PEP	NASDAQ
Pfizer Inc.	PFE	NYSE
The PNC Financial Services Group, Inc.	PNC	NYSE
PPG Industries, Inc.	PPG	NYSE
PPL Corporation	PPL	NYSE
QUALCOMM Incorporated	QCOM	NASDAQ
Quanta Services, Inc.	PWR	NYSE

Regions Financial Corporation	RF	NYSE
Roblox Corporation	RBLX	NYSE
Shell plc	SHEL	NYSE
Shopify Inc.	SHOP	NYSE
Sirius XM Holdings Inc.	SIRI	NASDAQ
Six Flags Entertainment Corporation	SIX	NYSE
Somnigroup International Inc.	SGI	NYSE
The Southern Company	SO	NYSE
Spotify Technology S.A.	SPOT	NYSE
State Street Corporation	STT	NYSE
Taiwan Semiconductor Manufacturing Company Limited	TSM	NYSE
Teradyne, Inc.	TER	NASDAQ
Tesla, Inc.	TSLA	NASDAQ
Uber Technologies, Inc.	UBER	NYSE
Under Armour, Inc.	UAA / UA	NYSE
Universal Corporation	UVV	NYSE
Verisk Analytics, Inc.	VRSK	NASDAQ
Vulcan Materials Company	VMC	NYSE
Walmart Inc.	WMT	NYSE
Warner Bros. Discovery, Inc.	WBD	NASDAQ
Webster Financial Corporation	WBS	NYSE
Wells Fargo & Company	WFC	NYSE
Westamerica Bancorporation	WABC	NASDAQ
Zions Bancorporation, N.A.	ZION	NASDAQ

Appendix E

Spearman's rho Correlation Table for AEO Variables and AI Accuracy

			Correlations					
			Overall_Score	Technical_SEO	Content_Structure	Schema_Markup	EEAT	Accuracy_AVG
Spearman's rho	Overall_Score	Correlation Coefficient	1.000	.361***	.842***	.227*	-.076	.400***
		Sig. (2-tailed)	.	<.001	<.001	.020	.439	<.001
Technical_SEO	Technical_SEO	N	105	105	105	105	105	105
		Correlation Coefficient	.361***	1.000	.452***	-.493***	-.738***	.238*
		Sig. (2-tailed)	<.001	.	<.001	<.001	<.001	.015
Content_Structure	Content_Structure	N	105	105	105	105	105	105
		Correlation Coefficient	.842***	.452***	1.000	-.174	-.310**	.347***
		Sig. (2-tailed)	<.001	<.001	.	.076	.001	<.001
Schema_Markup	Schema_Markup	N	105	105	105	105	105	105
		Correlation Coefficient	.227*	-.493***	-.174	1.000	.539***	.040
		Sig. (2-tailed)	.020	<.001	.076	.	<.001	.688
EEAT	EEAT	N	105	105	105	105	105	105
		Correlation Coefficient	-.076	-.738***	-.310**	.539***	1.000	-.022
		Sig. (2-tailed)	.439	<.001	.001	<.001	.	.825
Accuracy_AVG	Accuracy_AVG	N	105	105	105	105	105	105
		Correlation Coefficient	.400***	.238*	.347***	.040	-.022	1.000
		Sig. (2-tailed)	<.001	.015	<.001	.688	.825	.
		N	105	105	105	105	105	105

***. Correlation is significant at the 0.001 level (2-tailed).

*. Correlation is significant at the 0.05 level (2-tailed).

**. Correlation is significant at the 0.01 level (2-tailed).